Mock 02 solutions

SECTION A

Answer **ALL** questions in this section.

1. (a) Give the ϵ, N definition for the convergence of an infinite sequence, $\{x_n\}_{n=1}^{\infty}$, to a limit, L. Comment briefly on the roles played by the parameters ϵ and N in this definition. (5 marks)

Solution: The infinite sequence $\{x_n\}_{n=1}^{\infty}$ converges to a limit L if for every $\epsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n > N$, $|x_n - L| < \epsilon$.

The parameter ϵ represents the level of *closeness* required. We need to be able to make the difference between x_n and L smaller than any given ϵ.

The parameter N is the point in the sequence beyond which all terms are within ϵ of the limit L. The value of N depends on the choice of ϵ.

(b) State the definition, in terms of partial sums, of the convergence of an infinite series, $\sum_{n=1}^{\infty} x_n$. Then explain briefly why the *General Term Test for Divergence* is true, that if the sequence of terms $\{x_n\}_{n=1}^{\infty}$ does not converge to zero, the series must diverge. (5 marks)

Solution: An infinite series, $\sum_{n=1}^{\infty} x_n$, converges to a sum S if and only if the sequence of its partial sums, $\{S_k\}_{k=1}^{\infty}$, converges to S. That is, $\sum_{n=1}^{\infty} x_n = S$ if and only if $\lim_{k\to\infty} S_k = S$, where $S_k = \sum_{n=1}^{k} x_n$.

For the sequence of partial sums, S_k, to converge, they must be getting closer and closer to the limit S as $k \to \infty$. If the terms x_n converge to a non-zero value, or do not converge at all, then the partial sums will not stabilize and will continue to change significantly as more terms are added.

(c) Consider the sequence $\{x_n\}_{n=1}^{\infty}$ defined by $x_n = \frac{1}{2n-1}$. Using the ϵ, N definition of convergence, prove that $\lim_{n\to\infty} x_n = 0$. (8 marks)

Solution: Let $\epsilon > 0$ be given. We want to find $N \in \mathbb{N}$ such that for all $n > N$, $|x_n - 0| < \epsilon$. Since $x_n = \frac{1}{2n-1}$, we have $|x_n - 0| = \frac{1}{2n-1}$. Choose N such that $N > \frac{1}{2\epsilon} + \frac{1}{2}$. Then, for all $n > N$, $n > N > \frac{1}{2\epsilon} + \frac{1}{2}$, so $\frac{1}{2n-1} < \epsilon$. Therefore, by the definition, $\lim_{n\to\infty} x_n = 0$.

(d) Consider the series $\sum_{n=1}^{\infty} a_n$, where the term is defined by $a_n = \frac{3n^2+1}{2n^2}$. Use the general term test to prove that this series diverges.

(7 marks)

Solution: We consider the limit of the general term $a_n = \frac{3n^2+1}{2n^2}$ as n approaches infinity:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{3n^2+1}{2n^2} = \lim_{n\to\infty} \left(\frac{3+1/n^2}{2}\right) = \frac{3+0}{2} = \frac{3}{2}$$

Since $\lim_{n\to\infty} a_n = \frac{3}{2} \neq 0$, by the general term test for divergence, the series $\sum_{n=1}^{\infty} \frac{3n^2+1}{2n^2}$ diverges.

SECTION A

2. (a) Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous function. Using the concept of limiting value, give the formal definition of the derivative, $f'(a)$, of f evaluated at the point a.

Comment briefly on the geometric interpretation of the derivative, how it relates to the graph of the function f. (5 marks)

Solution: The derivative of a function f at a point a is defined as:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

provided this limit exists. Geometrically, the derivative at a point a represents the slope of the tangent line to the graph of the function f at that point. It indicates how steeply the function is increasing or decreasing at a.

(b) Give the definition of an anti-derivative, F, of a function, $f : \mathbb{R} \to \mathbb{R}$. How will two different anti-derivatives of f be related to each other? Justify your answer.

Solution: A function F is an anti-derivative of f if $F'(x) = f(x)$ for all $x \in \mathbb{R}$.

If F_1 and F_2 are two anti-derivatives of f, then by linearity $(F_1 - F_2)'(x) = f(x) - f(x) = 0$. Therefore, $F_1 - F_2$ must be a constant function (as it has a derivative of zero everywhere), and so $F_1 = F_2 + C$, for some real constant C.

(5 marks)

(c) Consider the function $g : \mathbb{R}\backslash\{0\} \to \mathbb{R}$ defined by $g(x) = \frac{1}{x}$. Using the definition of the derivative you gave in part (a), prove that $g'(x) = -\frac{1}{x^2}$.

(7 marks)

Solution: Using the definition of the derivative:

$$\begin{aligned} g'(x) &= \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} \\ &= \lim_{h\to 0} \frac{1/(x+h) - 1/xy}{h} \end{aligned}$$

$$
\begin{aligned}
&= \lim_{h \to 0} \left(\frac{1}{h} \frac{x - (x+h)}{(x+h)x} \right) \\
&= \lim_{h \to 0} \left(\frac{1}{h} \frac{-h}{(x+h)x} \right) \\
&= \lim_{h \to 0} \frac{-1}{(x^2 + hx)} \\
&= \frac{-1}{x^2}, \text{ by quotient and product rule for limits.}
\end{aligned}
$$

(d) Use the principle of anti-differentiation to find the unique function $h : \mathbb{R} \to \mathbb{R}$ that satisfies $h''(x) = -\sin(x)$ and has the initial values $h'(0) = 1$ and $h(0) = 0$. (8 marks)

Solution: First integrate $h''(x)$ to find $h'(x)$:

$$h'(x) = \int -\sin(x)dx = \cos(x) + C,$$

where C is a constant of integration. Next, we integrate $h'(x)$ to find $h(x)$:

$$h(x) = \int (\cos(x) + C)\, dx = \sin(x) + Cx + D,$$

where D is another constant of integration. Now apply initial conditions to find C and D.

$$h'(0) = \cos(0) + C = 1 + C = 1 \implies C = 0.$$

Then,

$$h(0) = \sin(0) + D = 0 + D = 0 \implies D = 0.$$

Therefore, the unique function h is:

$$h(x) = \sin(x).$$

END OF SECTION A

SECTION B

Answer **TWO** questions in this section.

3. (a) Let the sequence $\{c_m\}_{m=1}^{\infty}$ be defined by the formula

$$c_m = \frac{3m^4 + 2m^2 + 1}{-7m^4 + 8}.$$

Show how the algebra of limits theorem for convergent sequences can be used to prove that $\{c_m\}_{m=1}^{\infty}$ is a convergent sequence and to determine its limit. Make it clear in your answer where you are using each part of the theorem that you rely on. (7 marks)

Solution:

$$\begin{aligned}\lim_{m\to\infty} c_m &= \lim_{m\to\infty} \frac{3m^4 + 2m^2 + 1}{-7m^4 + 8} \\ &= \lim_{m\to\infty} \frac{3 + 2/m^2 + 1/m^4}{-7 + 8/m^4} \\ &= \frac{-3}{7},\end{aligned}$$

where the last step uses the quotient and linear combinations form of the algebra of limits theorem and the fact that $1/m^2, 1/m^4 \to 0$ as $m \to \infty$.

(b) Use the ϵ, N definition of sequence convergence to prove the product part of the algebra of limits theorem for convergent sequences. That is, prove that if $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ are convergent sequences with limits L and M respectively (with L and M assumed to be non-zero), then the *product* sequence $\{x_n y_n\}_{n=1}^{\infty}$ is a convergent sequence with limit LM. (7 marks)

Solution: Let $\epsilon > 0$ be given. Since $\{x_n\}_{n=1}^{\infty}$ converges to L and $\{y_n\}_{n=1}^{\infty}$ converges to M, there exist natural numbers N_1 and N_2 such that for all $n > N_1$, $|x_n - L| < \min\left(\sqrt{\frac{\epsilon}{3}}, \frac{\epsilon}{3|M|}\right)$, and for all $n > N_2$, $|y_n - M| < \min\left(\sqrt{\frac{\epsilon}{3}}, \frac{\epsilon}{3|L|}\right)$.
Let $N = \max(N_1, N_2)$. Then for all $n > N$, we have:

$$\begin{aligned}|x_n y_n - LM| &= |(x_n - L)(y_n - M) + L(y_n - M) + M(x_n - L)|, \text{ rewriting} \\ &\leq |x_n - L||y_n - M| + |L||y_n - M| + |M||x_n - L|, \text{ by the triangle inequality} \\ &< \sqrt{\frac{\epsilon}{3}}\sqrt{\frac{\epsilon}{3}} + |L|\frac{\epsilon}{3|L|} + |M|\frac{\epsilon}{3|M|}, \text{implementing upper-bounds}\end{aligned}$$

$$= \epsilon.$$

Thus, $\{x_n y_n\}_{n=1}^{\infty}$ converges to LM.

(c) Consider the sequence $\{a_n\}_{n=1}^{\infty}$ defined by $a_0 = 1$ and the recurrence relation

$$a_n = \frac{1}{5 - a_{n-1}}, \quad \text{for } n \geq 1.$$

In addition, assume that

- the sequence $\{a_n\}_{n=1}^{\infty}$ is bounded below by 0, and
- the sequence $\{a_n\}_{n=1}^{\infty}$ is monotone decreasing.

Use the Monotone Convergence Theorem to deduce that the sequence $\{a_n\}_{n=1}^{\infty}$ is a convergent sequence and then find its limit. Explain your reasoning carefully and point out any properties of sequence convergence that you use. (7 marks)

Solution: Since we are given that the sequence is bounded below by 0 and is monotone decreasing, we can conclude that it converges, by the Monotone Convergence Theorem. Let L be the limit of the sequence. Since a_n is defined by the recurrence relation $a_n = \frac{1}{5-a_{n-1}}$, we can find the limit by taking the limit on both sides as $n \to \infty$:

$$L = \frac{1}{5 - L}.$$

Note $L \neq 5$, as $a_0 = 1$ and the sequence decreases. Multiplying both sides by $5 - L$, we get:

$$5L - L^2 = 1.$$

Rearranging gives us the quadratic equation:

$$L^2 - 5L + 1 = 0.$$

which has the solutions

$$L = \frac{5 \pm \sqrt{21}}{2}.$$

Note that the larger solution satisfies $\frac{5+\sqrt{21}}{2} > 1$. But the sequence decreases from $a_0 = 1$, therefore the limit L must be the smaller solution, so

$$L = \frac{5 - \sqrt{21}}{2}.$$

Thus, the limit of the sequence $\{a_n\}_{n=1}^{\infty}$ is $\frac{5-\sqrt{21}}{2}$.

(d) Prove the *monotone decreasing* assumption from part (c). (4 marks)

Solution: Use induction. Observe that $a_0 = 1$ and $a_1 = 1/4$. So we can say as a base case that

$$5 > a_0 > a_1 > 0.$$

Now assume that

$$5 > a_{k-1} > a_k > 0,$$

for some $k \geq 1$. Using properties of inequalities and the induction assumption we can argue that

$$\begin{aligned}
5 > a_{k-1} > a_k > 0 &\Rightarrow -5 < -a_{k-1} < -a_k < 0 \\
&\Rightarrow 0 < 5 - a_{k-1} < 5 - a_k < 5 \\
&\Rightarrow \frac{1}{5 - a_{k-1}} > \frac{1}{5 - a_k} > 0 \\
&\Rightarrow 5 > a_k > a_{k+1} > 0.
\end{aligned}$$

Therefore, by induction, for all $n \geq 0$ we have $a_n > a_{n+1}$, i.e. the sequence is monotonically decreasing.

SECTION B

4. (a) Consider the infinite series

$$\sum_{n=1}^{\infty} \frac{-2+3^n}{5^{n+3}}.$$

Use the result about the convergence of geometric series to prove that the series defined above is convergent and determine its sum. You can use the theorem about linear combinations of convergent series. (8 marks)

Solution: We can split the partial sum up into the sum of two summations, which are both partial sums of geometric series. Then apply the result on convergence and limits of geometric series to the sum of the original series as follow:

$$\begin{aligned}\sum_{n=1}^{k} \frac{-2+3^n}{5^{n+3}} &= \sum_{n=1}^{k} \left(\frac{-2}{5^4}\left(\frac{1}{5}\right)^{n-1}\right) + \sum_{n=1}^{k} \left(\frac{3}{5^4}\left(\frac{3}{5}\right)^{n-1}\right), \\ &\to \frac{-2/5^4}{1-1/5} + \frac{3/5^4}{1-3/5}, \text{ as } k \to \infty \\ &= \frac{-1}{2\cdot 5^3} + \frac{3}{2\cdot 5^3} \\ &= \frac{1}{125}.\end{aligned}$$

(b) Use a suitable comparison test to prove that the series

$$\sum_{n=1}^{\infty} \frac{3n^2}{9-n^4},$$

is a convergent series. You can make use of the known convergence of any hyper-harmonic series. (5 marks)

Solution: We compare to the convergent series $\sum 1/n^2$, using the limit comparison test. The limit to examine is

$$\lim_{n\to\infty}\left(\frac{3n^2}{9-n^4}\cdot n^2\right) = \lim_{n\to\infty}\frac{3n^4}{9-n^4}$$

$$
\begin{aligned}
&= \lim_{n\to\infty} \frac{3}{9/n^4 - 1}\\
&= -3, \text{ since } 9/n^4 \to 0 \text{ as } n \to \infty.
\end{aligned}
$$

Therefore by the limit comparison test and the fact that $\sum 1/n^2$ is a convergent hyper-harmonic series, the original series converges also.

(c) Use the ratio test to prove that the series

$$\sum_{m=1}^{\infty} \frac{5^m}{(m+3)!},$$

is a convergent series. *Note that* ! *denotes the factorial operation*.

(5 marks)

Solution: Taking the limit of the ratio of consecutive terms $x_m = \frac{5^m}{(m+3)!}$ we see

$$
\begin{aligned}
\lim_{m\to\infty} \frac{x_{m+1}}{x_m} &= \lim_{m\to\infty} \left(\frac{5^{m+1}}{(m+4)!} \frac{(m+3)!}{5^m} \right)\\
&= \lim_{m\to\infty} \left(\frac{5}{m+4} \right)\\
&= 0, \text{ by algebra of limits theorem.}
\end{aligned}
$$

Since the limit is 0, the original series converges by the ratio test as the limit of the ratio of consecutive terms has absolute value less than 1.

(d) Obtain the partial fraction expansion of

$$\frac{21}{m^2 + 3m},$$

and use it to obtain the sum of the infinite series

$$\sum_{m=1}^{\infty} \frac{21}{m^2 + 3m},$$

by considering cancellation effects in the partial sums of this series.

(7 marks)

Solution: The required partial fraction expansion is

$$\frac{21}{m^2+3m} = \frac{7}{m} - \frac{7}{m+3},$$

so in the partial sum we get cancellation of all but three terms at the beginning and three at the end,

$$\begin{aligned} S_k &= \sum_{m=1}^{k} \frac{21}{m^2+3m}, \\ &= \sum_{m=1}^{k} \left(\frac{7}{m} - \frac{7}{m+3} \right), \\ &= \frac{7}{1} + \frac{7}{2} + \frac{7}{3} - \frac{7}{k+1} - \frac{7}{k+2} - \frac{7}{k+3}. \end{aligned}$$

So

$$\sum_{m=1}^{\infty} \frac{21}{m^2+3m} = \lim_{k\to\infty} S_k = 77/6,$$

by applying the algebra of limits theorem to the limit of S_k.

SECTION B

5. (a) Use the limit definition of the derivative to prove the product rule for differentiation. That is, prove that if f and g are differentiable functions, then the derivative of their product is given by

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Point out clearly where you make use of any properties of the convergence of functions. (5 marks)

Solution: Let f and g be differentiable at x. Consider the difference quotient

$$\frac{(fg)(x+h)-(fg)(x)}{h} = \frac{f(x+h)g(x+h)-f(x)g(x)}{h} = \dots$$

Add and subtract $f(x)g(x+h)$ to obtain the algebraic split

$$\dots = \frac{(f(x+h)-f(x))}{h}g(x+h) \ + \ f(x)\,\frac{g(x+h)-g(x)}{h}.$$

Since differentiability implies continuity, $g(x+h) \to g(x)$ as $h \to 0$. Taking limits and using linearity and product rules for limits gives

$$\begin{aligned}&\lim_{h\to 0}\frac{(fg)(x+h)-(fg)(x)}{h}\\&= \left(\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}\right)g(x) + f(x)\left(\lim_{h\to 0}\frac{g(x+h)-g(x)}{h}\right).\end{aligned}$$

Hence, by the limit definition of the derivative,

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x),$$

as required.

(b) Consider the function g defined by the formula

$$g(x) = 2x^4\cos(3x^2).$$

Use the linearity, product and chain rules to obtain the first and second order derivatives, $g'(x)$ and $g''(x)$, of g. Point out clearly where you are using each of the rules. (5 marks)

Solution: Using the product rule and chain rules we get

$$g'(x) = 8x^3\cos(3x^2) - 2x^4\sin(3x^2)6x = -12\,x^5\sin\left(3\,x^2\right) + 8\,x^3\cos\left(3\,x^2\right).$$

Then using product and chain rules again we get, after simplification,

$$g''(x) = -72\,x^6\cos\left(3\,x^2\right) - 108\,x^4\sin\left(3\,x^2\right) + 24\,x^2\cos\left(3\,x^2\right).$$

(c) Consider the graph in the xy-plane defined by the equation $y = \sqrt{x}$, for $x \geq 0$. Use the concepts of *differentiation* and *stationary point* to find the point on this curve that is closest to the point $(5, 0)$. (10 marks)

Solution: The distance of a point $(x, y) = (x, \sqrt{x})$ on the curve from the point $(5, 0)$ is given by

$$D(x) = \sqrt{(x-5)^2 + \sqrt{x}^2} = \sqrt{x^2 - 9x + 25}.$$

The stationary point(s) of D are the solution(s) to $D'(x) = 0$, i.e.

$$D'(x) = \frac{1}{2}(x^2 - 9x + 25)^{-1/2}(2x - 9) = 0.$$

This has the unique solutions $x = 9/2$, (from the zero of the numerator factor $(2x - 9)$).

That this is a local minimum can be justified by the fact that there is clearly no maximum possible distance as points can be found on the curve arbitrarily far from $(5, 0)$ by choosing sufficiently large x. Alternatively we can apply a second-order derivative test.

So the point on the curve closest to $(5, 0)$ is the point $(9/2, \sqrt{9/2})$.

(d) Use a combination of the product rule and quotient rule to fully expand the derivative F', of a function F defined by

$$F(x) = \frac{u(x)}{v(x)w(x)z(x)},$$

and show how it depends on the individual functions u, v, w, z and their derivatives. You can assume that v, w and z are always non-zero.

(5 marks)

Solution: Applying the quotient rule gives

$$F'(x) = \frac{(vwz)(x)u'(x) - (u)(x)(vwz)'(x)}{(vwz)^2(x)}$$

and then the product rule across the second term in the numerator gives

$$\begin{aligned}
F'(x) &= \frac{u'(x)v(x)w(x)z(x)}{(vwz)^2(x)} - \frac{u(x)\Big(v'(x)w(x)z(x) + v(x)w'(x)z(x) + v(x)w(x)z'(x)\Big)}{(vwz)^2(x)} \\
&= \frac{u'(x)v(x)w(x)z(x) - u(x)v'(x)w(x)z(x) - u(x)v(x)w'(x)z(x) - u(x)v(x)w(x)z'(x)}{v^2(x)w^2(x)z^2(x)}
\end{aligned}$$

which is in the required form.

SECTION B

6. (a) Show how integration by parts can be applied to evaluate the definite integral

$$I_1 = \int_{-3\pi/2}^{3\pi/2} e^{2x} x^2 \, dx.$$

(6 marks)

Solution: This requires integration by parts twice,

$$\begin{aligned}
I_1 &= \left[\frac{e^{2x}}{2} x^2\right]_{-3\pi/2}^{3\pi/2} - \int_{-3\pi/2}^{3\pi/2} e^{2x} x \, dx, \\
&= \left[\frac{e^{2x}}{2} x^2\right]_{-3\pi/2}^{3\pi/2} - \left[\frac{e^{2x}}{2} x\right]_{-3\pi/2}^{3\pi/2} + \frac{1}{2}\int_{-3\pi/2}^{3\pi/2} e^{2x} \, dx \\
&= \left[\frac{e^{2x}}{2} x^2\right]_{-3\pi/2}^{3\pi/2} - \left[\frac{e^{2x}}{2} x\right]_{-3\pi/2}^{3\pi/2} + \left[\frac{e^{2x}}{4}\right]_{-3\pi/2}^{3\pi/2}
\end{aligned}$$

which after evaluation at the limits simplifies to

$$I_1 = \frac{1}{8}\left(-6\,\pi + 9\,\pi^2 + 2\right) e^{3\,\pi} - \frac{1}{8}\left(6\,\pi + 9\,\pi^2 + 2\right) e^{-3\,\pi}.$$

(b) Use the substitution $u = x + 5$ to evaluate the indefinite integral

$$I_2 = \int \frac{2x^2 + 3x + 1}{x + 5} \, dx.$$

(6 marks)

Solution: By substitution, using $u = x + 5$, i.e. $x = u - 5$, we have $dx = du$ and the integral becomes.

$$\begin{aligned}
I_2 &= \int \frac{2(u-5)^2 + 3(u-5) + 1}{u} \, du \\
&= \int \frac{2\,u^2 - 17\,u + 36}{u} \, du \\
&= \int 2u - 17 + \frac{36}{u} \, du \\
&= u^2 - 17u + 36 \ln(|u|) + C,
\end{aligned}$$

$$= (x+5)^2 - 17(x+5) + 36\ln(|x+5|) + C,$$

for some constant C.

(c) The Riemann integral of a function $f(x)$ between the limits $x = a$ and $x = b$ is defined as

$$\int_a^b f(x)\,dx = \lim_{\substack{n\to\infty \\ \Delta x\to 0}} \left(\sum_{i=1}^{n} f(x_i^*)\Delta x_{i-1}, \right).$$

Set up and evaluate the limit in the Riemann integral to show that

$$\int_0^5 x^3\,dx = \frac{625}{4}.$$

In your answer you should clearly define the partition of the interval $(0, 2)$ you are using and what the objects $x_i^*, \Delta x_{i-1}$ and Δx are, and point out any properties of the limit process you use.

You can use, without proof, the summation formula

$$\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}.$$

(13 marks)

Solution: We will use the equally spaced partition $\{x_i\}_{i=1}^{n+1}$ defined by

$$x_i = (i-1)\frac{5}{n}.$$

For the representative point x_i^* we shall use the left-edge of each sub-interval, i.e.

$$x_i^* = (i-1)\frac{5}{n}, \quad \text{for } i = 1, \ldots, n.$$

Each sub-interval width is the same at $\Delta x_{i-1} = \frac{2}{n}$ and so Δx, the maximum sub-interval width also is $\Delta x = \frac{5}{n}$.

$$I = \int_0^5 x^3\,dx = \lim_{n\to\infty} \left(\sum_{i=1}^{n} \left((i-1)\frac{5}{n} \right)^3 \frac{5}{n} \right),$$

$$
\begin{aligned}
I &= \lim_{n\to\infty}\left(\sum_{i=1}^{n}\frac{5^4}{n^4}\left((i-1)^3\right)\right),\\
&= 625\lim_{n\to\infty}\left(\frac{1}{n^4}\sum_{i=1}^{n-1} i^3\right),\\
&= 625\lim_{n\to\infty}\left(\frac{1}{n^4}\frac{(n-1)^2(n^2)}{4}\right),\\
&= 625\lim_{n\to\infty}\left(\frac{1}{n^4}\frac{n^4-2n^3+n^2}{4}\right),\\
&= \frac{625}{4}\lim_{n\to\infty}\left(1-2/n+1/n^2\right),\\
&= \frac{625}{4}
\end{aligned}
$$

as required.

END OF SECTION B
END OF QUESTIONS